

**SECTION 01000-A
GENERAL TECHNICAL REQUIREMENTS**

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2.0 SITE DESIGN CRITERIA

- A. Provide equipment, components, structures, and paving in accordance with the following design criteria unless otherwise specified:
1. Location: Tainan Industrial Park.
 - a. County: Tainan.
 - b. Locality: Annan Area.
 2. Seismic Criteria:

a. Seismic for structures design: Taiwan Building Code, 2022 edition:

1) Site Design Spectral Response:

a) Site Design Spectral Response

Acceleration at Short Period, $S_s^D=0.7$

b) Site Design Spectral Response

Acceleration at One Second Period, $S_1^D=0.40$

c) Site Maximum Design Spectral Response

Acceleration at Short Period, $S_s^M=0.9$

d) Site Max. Design Spectral Response

Acceleration at One Second Period, $S_1^M=0.50$

~~2) Near-fault factor $N_A=1.0$, $N_V=1.0$~~

3) Importance factor:

a) $I=1.0$ for Fab, Support, CUP.

b) $I=1.0$ for Gas Yard.

c) $I=1.0$ for office, warehouse.

b. Seismic for CR and M/E/P seismic bracing:

Seismic Resistance Requirements - All equipment, anchorage and bracing, hangers and bracing for cable trays and pipes shall be designed in accordance with the Taiwan Building Code, 2022 edition.

1) $F_{pH} = 0.4 S_{DS} I_p (a_p/R_p) (1 + 2 h_x/h_n) W_p$

a) $F_{pmax} = 1.6 S_{DS} I_p W_p$

b) $F_{pmin} = 0.3 S_{DS} I_p W_p$

~~c) $\overline{V_s}_{30} > 270$ m/s soil profile type=1~~

2) $F_{pv} = 1/2 F_{ph}$

3) Terms in the calculations:

a_p = Component amplification factor that varies from 1.00 to 2.50 (select appropriate value from Table 4-1 or 4-2).

F_p = Seismic design force centered at the component's center of gravity and distributed relative to component's mass distribution.

I_p = Component importance factor that is 1.0, as determined in Section 4.2.

h_n = Average roof height of structure relative to the base elevation.

R_p = Component response modification factor that varies from 1.0 to 3.5 (select appropriate value from Table 4-1 to 4-2).

S_{DS} = Design spectral response acceleration at short period, as determined in Section 2.4.

W_p = Component operating weight.

h_x = Height in structure at point of attachment of component. For items at or below base, h_x shall be taken as zero. For items at or above the roof, h_x is not required to be taken as greater than the roof height h .

4) Value of mechanical and electrical coefficients as defined in TBC 2022 Table 4-2 are shown on the following table:

Mechanical and Electrical Component or Element	a_p	R_p
1. General Mechanical		
a. Boilers and Furnaces	1.0	2.5
b. Pressure Vessels on Skirts and Free-Standing	2.5	2.5
c. Stacks	2.5	2.5
d. Cantilevered Chimneys	2.5	2.5
e. Other	1.0	2.5
2. Manufacturing and Process Machinery		
a. General	1.0	2.5
b. Conveyors (Nonpersonnel)	2.5	2.5
3. Piping Systems: Limited Deformation Elements and Attachments	1.0	2.5
4. HVAC System Equipment		

Mechanical and Electrical Component or Element	a _p	R _p
a. Vibration Isolated	2.5	2.5
a. Nonvibration Isolated	1.0	2.5
b. Mounted In-Line with Ductwork	1.0	2.5
c. Other	1.0	2.5
5. Elevator Components	1.0	2.5
6. Escalator Components	1.0	2.5
7. General Electrical		
a. Distributed Systems (Bus Ducts, Conduits, Cable Trays)	1.0	3.5
b. Equipment	1.0	2.5
8. Lighting Fixtures	1.0	1.25

3. Temperature (Outdoors):

Area	Summer	Winter
Critical areas	35°C dry bulb 80% Relative Humidity (RH)	5°C dry bulb 50% RH 2.7gw/kg
Non-critical areas	33°C dry bulb (1) 27°C wet bulb (1)	8°C dry bulb (2) 50% RH (estimate)
Cooling Towers	29.4°C wet bulb	

Notes for Outside Air Design conditions table:

(1) ASHRAE 1%, Tainan

(2) ASHRAE 99%, Tainan

4. Wind Loading: per Taiwan Building Code 2015, Edition, I_p =1.1.

3.0 DEFINITIONS

A. Owner:

Micron Memory Taiwan Co., Ltd.

B. Construction Manager:

1. Owner will provide construction management services themselves through person of the contact.
2. Duties and responsibilities of the Construction Manager with regard to the Contract Document are as follows: The Construction Manager is appointed as the Owner's representative for work performed under this Contract. Where the Specifications require Owner review, authorization, approval, investigation, concurrence, or other action the Construction Manager shall provide such action. Where the Specifications require Contractor to submit to, notify, report to, cooperate with, or otherwise provide information to the Owner, the Contractor shall provide such information to the Construction Manager. In general the rights and obligations of the Owner under this Contract will be administered by the Construction Manager.

C. Vendor: The authorized distributor or agent of the manufacturer to make direct sale of the specified products and entering into an agreement with Owner. Wherever the word "Contractor" appears in Division 1 through Division 16 of the Specifications as it relates to the furnishing, it shall be synonymous with "Vendor" unless otherwise defined.

D. Changes: Modifications to the Contract.

1. Change Orders: Changes to the scope of work which impact the Contract sum or Contract time require a fully executed change order.
2. Construction Change Directive: written order prepared by the Engineer and signed by the Owner directing Contractor to make changes to the scope of work which may or may not impact Contract sum or Contract time, but is evidence that the parties expect that the construction change directive will be incorporated into a subsequently issued change order.
3. Field Orders: Changes to the scope of work not requiring change in Contract sum or Contract time will be made by a field order. If Contractor believes a field order impacts Contract sum or

Contract time, Contractor shall notify Owner in writing within 3 calendar days and before starting the field order change, except in an emergency.

- E. Hazardous and toxic materials shall mean hazardous substances as defined in the National Fire Protection Agency. Hazardous and toxic waste as defined in the Taiwan Environmental Protection Agency regulation requirements and similar terms as used in applicable national and local statutes, rules, and regulations to include carcinogenic, acutely toxic, flammable, or explosive materials in any form.

4.0 CORRESPONDENCE

- A. Communications:
 - 1. Direct communications, correspondence, and submittals for the project to the Owner at the address above.
 - 2. Maintain log of communications and correspondence.
 - 3. Limit letters or faxes to a single subject and identify by a transmittal number correlated to the correspondence log.
 - 4. Confirm telephone conversations in writing.
 - 5. Progress Reports: Provide monthly report to include, as a minimum, the following.
 - a. Activity during past month.
 - b. Confirmation of schedule.
- B. Language: Prepare and submit Contract Documents including but not limited to submittals, correspondence, requests for information(RFIs), and operation and maintenance(O & M) manuals in Chinese. If otherwise required by the Contract prepare a translated copy in the language of the country where the project is located. To the maximum extent permitted by the governing law of the Contract, the Chinese version shall govern if there is any inconsistency or conflict between the two translations.

5.0 MANUFACTURER'S REPRESENTATIVE REQUIREMENTS

- A. Specifications may require a manufacturer's representative be provided for specified functions on site for a stated period of time. The daily rate to be charged for providing the manufacturer's representative beyond the stated period of time as may be required to complete the specified activities is shown in the Contract. The daily rate for the additional services shall be the same rate that will be used to reduce the

Contract sum to account for fewer days spent than the number stated in the Specifications.

- B. During subsequent work activities by the installing workers, the manufacturer's representative shall provide periodic, timely inspection and counseling to ensure system complies with factory requirements and the specifications and that it will operate at the specified level of performance.
- C. The manufacturer's representative shall submit a written report listing tasks performed, observations, and finished results.

6.0 CONFLICTS WITH OR WITHIN THE CONTRACT DOCUMENTS

- A. If, during the performance of the work, the Contractor discovers conflict, error, ambiguity, or discrepancy within the Contract Documents, or between the Contract Documents and provisions of laws or regulations applicable to the performance of the work, or of standards, specifications, manuals, codes, or of instructions of manufacturers or vendors, notify the Owner in writing immediately. Do not proceed with the work affected by the conflict (except in an emergency) until an amendment or supplement to the Contract Documents has been issued.

7.0 SUBMITTAL PROCEDURES

- A. General:
 - 1. Submittals shall be accompanied by a transmittal that lists all attachments.
 - 2. Review of submittals by the Engineer will be for the purpose of checking for conformance with information given and the design concept expressed in the Contract Documents, and shall not relieve Contractor from responsibility for errors or omissions contained in the submittals. Engineer may make changes or corrections to the submittals, reproduce copies for its own use and return a marked up copy to the Contractor. If major revisions are required, the submittal will be returned to the Contractor to correct and resubmit. If Contractor proceeds with ordering materials or equipment without a completed Engineer review, it does so at its sole risk.
 - 3. Submittals will be acted upon by the Engineer as promptly as possible and returned to the Contractor no later than 10 working days after receipt.
 - 4. Mark submittals with the following, as a minimum:

- a. Specification section number.
 - b. Project identification.
 - c. Purchase order number, if issued.
 - d. Owner's tag or identification number for products, instruments, and equipment.
 - e. Line number(s) on piping drawings.
 - f. Date of issue.
 5. Revisions to documents previously submitted shall be clearly identified as to which items specifically were revised.
 6. The quantity of submittals required, at a minimum, is:
 - a. Product Data (Submittals That can be Duplicated on a Copy Machine, Usually A4 Size): six copies.
 - b. Shop Drawings: one reproducible sepia and three blue lines.
 - c. Samples: three sets.
 7. The practice of submitting incomplete or unchecked submittals is not acceptable. Submittals which, in the opinion of the Engineer, clearly indicate that they have not been stamped, dated, and checked by Contractor will be considered as not complying with the intent of the Contract Documents and will be returned to Contractor for resubmission in the proper form.
- B. Bid Submittals: Submit the following with the Bid.
1. Exceptions to the Bid document requirements.
 2. Preliminary project schedule showing equipment delivery and work completion times, starting with execution of agreement.
 3. Specific condition or delivery time period for equipment or special material that may adversely affect the stated project completion and equipment delivery time period.
 4. List of proposed manufacturers for the primary components.
 5. Product handling, storage, assembly, installation, or adjustments required that impact design or installation of other work.
 6. List of substitutions proposed by the bidder.
 7. List of itemized alternates requested.
 8. List of spare parts with delivery time longer than 1 month.
 9. Product data for equipment and controls used on specified equipment, to include catalog sheets and specifications for proposed products.
 10. Contractor's construction safety and health plan.

11. Identify hazardous substances, as defined in paragraph 1.3 above that will be provided in any form with deliverables.
- C. Submittals During Construction Period: Within the required number of weeks after award of Contract, provide specified shop drawings, diagrams, illustrations, schedules, catalog cuts, design data, calculations, and other data specifically prepared for the project work by Contractor or a subcontractor, manufacturer, supplier, vendor, or distributor and the site-specific safety plan, hazardous waste management control plan, waste management plan, and proposed site layout.
1. As-Built Shop Drawings: Submit one set of reproducible, as-built shop drawings and one set of AutoCAD disks with shipment of the equipment.
 2. Independent Testing and Inspection Reports: Submit one copy of test or inspection results, for review by the Engineer, within 48 hours of the test or inspection. If the test failed, indicate on a subsequent report the action taken to clearly show how the deficiency was corrected.
 3. Equipment Submittals: Provide the following with each equipment submittal.
 - a. Proposed methods of attachments.
 - b. Supported weights and details for attachment to the structure.
 - c. Supported weights and details for loads greater than specified attached to floor joists, roof joists, and girders.
- D. Submittals Before Substantial Completion:
1. Operating and Maintenance Manuals: Provide six copies of the manuals within sufficient time to allow for training of Owner's personnel. Provide in three-ring, side-entry binders with durable plastic covers containing a detailed table of contents with tab dividers for major sections and special equipment and including:
 - a. Description of the system provided.
 - b. Handling, storage, and installation instructions.
 - c. Detailed description of the function of each principal component of the system or equipment.
 - d. Operating Procedures:
 - 1) Prestartup activities required.
 - 2) Startup.
 - 3) Normal operation.
 - 4) Emergency shutdown.
 - 5) Normal shutdown.

- 6) Troubleshooting guide.
- e. Maintenance:
 - 1) Complete Lubrication Requirements: type and source of lubricant, interval between lubrication, etc.
 - 2) Preventative and repair maintenance procedures.
 - 3) Complete spare parts list with cross-reference to original equipment manufacturer's part numbers.
- f. Control and Alarm Features:
 - 1) A schematic of control systems.
 - 2) Control loop electric ladder or functional block diagrams.
 - 3) A listing of controller operating set points.
 - 4) A listing of settings for alarms and shutdown systems.
- g. Safety and environmental considerations.

2. Executed warranty documents.

E. Cost of Submittals:

- 1. Preparation, assembly, review, and acceptance of substitutions, schedules, shop drawings, lists of materials, samples, mock-ups, and procedures submitted by Contractor shall be included in the Contract sum.
- 2. Additional costs, which may result from providing submittals, shall be solely the obligation of Contractor.

8.0 CODES AND STANDARDS

- A. Comply with the latest codes and other legal requirements applicable to the materials, equipment, and services provided in accordance with the Contract, even if an applicable code or legal requirement is not referenced in the project specifications.
- B. Promptly submit written notice to Engineer of observed variances between the Contract Documents and legal requirements.
- C. Materials, fasteners, piping systems, and equipment shall conform to ANSI and other U.S. standard dimensional requirements unless otherwise specifically required by the Specifications.
- D. Comply with the latest local and international codes and standards, including the following:

Local Codes and Standards

Design Criteria: Comply with local codes, including:

- Taiwan Science-based Industrial Park Standards
- Taiwan Building Code, 2011
- Taiwan Fire Code, 2013
- Taiwan Safety Code, 2013
- Taiwan EPA Regulations
- Taiwan Power Regulations

U.S. Codes and Standards

Design Criteria: U.S. codes and standards used as design guidelines:

- 2013 NFPA 72, National Fire Alarm Code
- 2015 NFPA 318, Standard for Protection of Cleanrooms
- ANSI B31.3, Standard for Process Piping
- National Electric Code (NEC)
- National Fire Protection Code
- American Society of Testing and Materials (ASTM)
- American National Standards Institute (ANSI)
- Underwriters Laboratories (UL)
- Factory Mutual (FM)
- American Society of Heating, Refrigeration and Air Conditioning Engineers (ASHRAE)
- Federal Standard 209E
- Institute of Environmental Sciences (IES)
- Sheet Metal and Air Conditioning National Association (SMACNA)
- Air Movement and Control Association (AMCA)
- Associated Air Balance Council (AABC)

- Instrument Society of America (ISA)

Exceptions to U.S. codes and standards:

- Conflicts between U.S. and Taiwan codes: review each conflict with Owner to obtain a decision on which code to use. The standard will be Taiwan Building Code
- Accessibility for disabled persons: Comply with Taiwan Building Code, 2011
- Maximum distance to exit:

Fabrication Building,
Administration/Support/Warehouse, and Central
Utility Building: 70 meters, as required by Taiwan Building Code, 2011
- Fire ratings for building structure (columns, beams, girders, floor and roof decks): Comply with Taiwan Building Code, 2011
- Door fire ratings: Comply with Taiwan Building Code, 2011
- Steel Trusses: 2-hour and 1-hour rated thin-film intumescent coating on trusses, steel beams and columns; as indicated on drawings
- Duct Penetrations through Fire Rated Walls: Alternate means of protection required

Owner Safety, Health, Environmental Protection Department

Design Criteria: Comply with the following standards:

- NFPA 1: Fire Prevention Code
- NFPA 12: Carbon Dioxide Extinguishing System
- NFPA 13: Standard for the Installation of Sprinkler System
- NFPA 15: Water Spray Fixed System For the Fire Protection
- NFPA 16: Deluge Form-water System
- NFPA 22: Water Tanks
- NFPA 30: Flammable and Combustible Liquids Code

- NFPA 50: Bulk Oxygen System
- NFPA 50A: Gaseous Hydrogen System
- NFPA 55: Compressed and Liquefied Gases in Portable Cylinders
- NFPA 68: Venting of Deflagrations
- NFPA 69: Explosion Prevention System
- NFPA 70E: Electrical Safety in Employee Work
- NFPA 72: National Fire Alarm Code
- NFPA 80: Fire Doors and Fire Windows
- NFPA 80A: Exterior Fire Exposure
- NFPA 92A: Recommended Practice for Smoke Control System
- NFPA 101: Code for Safety to Life from Fire I Buildings and Structures
- NFPA 105: Smoke-Control Door Assemblies
- NFPA 110: Emergency Standby Power System
- NFPA 170: Fire Safety Symbols
- NFPA 204m: Smoke, Heat Venting
- NFPA 214: Water Cooling Towers
- NFPA 220: Type Building Construction
- NFPA 231: General Storage
- NFPA 231C: Rack Storage of Materials
- NFPA 291: Fire Hydrants
- NFPA 318: Standard for the Protection of Clean Rooms
- NFPA 325: Prop of Flammable Liquids, Gases, Solids
- NFPA 497: Flammable Liquid Gases or Vapors and of Hazardous Location for Electrical Inst
- NFPA 820: Wastewater Facilities

- NFPA 1420: Warehouse Occupancies
- NFPA 2001: Standard on Clean Agent Fire Extinguishing System
- OSHA Code
- FM 1-6: Cooling Tower
- FM 1-44: Damage Limiting Construction (DLC)
- FM 7-7 and FM 7-12: Semiconductor Fabrication Facilities
- FM 7-45: Instrumentation and Control in Safety Applications
- IEC 61508: Functional Safety of Electrical/Electronic/Programmable Electronic Safety-Related Systems
- IEC 61511: Functional Safety - Safety Instrumented Systems for the Process Industry Sector
- EEUMA Publication No. 191: Alarm Systems - A Guide to Design, Management and Procurement

9.0 PRODUCT REQUIREMENTS

A. General:

1. Products shall comply with the project specifications, referenced standards, and applicable regulatory requirements.
2. Like items of equipment and distinct materials provided shall be from one manufacturer in order to achieve standardization for appearance, maintenance, and replacement.
3. Provide manufacturer's standard finish and color unless specified otherwise.
4. Products shall be new and suitable for the intended use.
5. Do not install previously used products, except as specifically required or allowed by Contract Documents.
6. The use of asbestos or other hazardous substances in any product is prohibited unless specifically approved by Owner.

B. Product Options:

1. Products specified by referenced standards or by description only shall meet the standards and applicable requirements of the Contract Documents.
2. Where products are specified by brand name or model number with one or more listed acceptable manufacturers but is not classified as sole source, products from other manufacturers may be considered for substitution. Submit requests for substitution in accordance with the requirements specified below.

C. Substitutions:

1. Provide each request in writing to the Engineer with complete data substantiating compliance of proposed substitution with the Contract Documents.
2. Substitution requests may be submitted either before the Bid or any time during construction. Prebid approval of substitute products is not required. Requests for substitutes during the Bid period will not be acted upon by the Engineer prior to Bid. Submit Bid prices based only on approved products.
3. Request for substitution constitutes a representation that Contractor:
 - a. Has investigated proposed product and determined that it meets or exceeds the specified product.
 - b. Shall provide the same warranty for the substitution as for the specified product.
 - c. Shall coordinate installation and make other changes that may be required for work to be complete.
 - d. Waives claims for additional costs that may subsequently become apparent due to the substitution.
4. Engineer will determine acceptability of proposed substitution and, except for requests submitted prior to Bid, will notify Contractor of acceptance or rejection in writing within 10 working days of receipt of the written request.
5. The use of substitute materials is not allowed without prior written approval by the Engineer.

D. Alternates:

1. Alternates are defined as alternate products, materials, equipment, or systems for work that may, at the Owner's option and under the terms established by the Instructions to Bidders, be selected and recorded in this Contract to either supplement or displace corresponding basic requirements of the project specifications. Alternates may or may not substantially change the scope and general character

of work and shall not be confused with allowances, prices, change orders, substitutions, and other similar provisions.

2. Coordinate alternate work with related work to ensure that each accepted alternate is completely integrated and interfaced with related work.
3. Within 2 weeks of award of Contract, the Owner will prepare and distribute to each contractor to be involved in the project a notification of status of each alternate. This notice will indicate which alternates have been accepted, rejected, and deferred for consideration at a later date. The notice will also include a full description of negotiated modifications to alternates, if any.

E. Delivery, Storage, and Handling:

1. Package and transport products by methods to avoid damage during shipment and storage prior to installation. Deliver in undamaged condition in manufacturer's unopened containers or packaging, unload and store in a dry and clean condition. Take special care to protect products that may become pitted, rusted, corroded, or which may deteriorate during shipment or storage. Machinery shall have a wrapping of heavy-duty weatherproof plastic. Protect sealed interiors from moisture damage through the use of silica gel or similar desiccants. Fragile components, instruments, controls, and other delicate parts shall be packaged separately to reduce the possibility of damage during shipment and shall be conspicuously marked "FRAGILE". Handling and storage instructions shall accompany the shipment.
2. Handle products as recommended by the manufacturer with approved equipment and in a manner that will prevent damage. Clearly mark sling posts, lifting lugs, or other material handling devices. Products shall be suitably crated, skidded, boxed, sealed, or otherwise protected and shall be properly braced against movement during shipment. Instructions for removal of bracing shall accompany the shipment.
3. Provide adequate weather protection such as waterproof tarpaulins, dunnage, or platforms above ground level and housings as required to protect the equipment and materials from weather. Provide temporary heating as necessary to protect covered equipment such as electrical gear from condensation where such could cause damage.
4. Inspect shipments promptly to ensure that products comply with requirements, quantities are correct, and

products are undamaged. Notify the Owner of evident discrepancies or damage.

10.0 SUPPORTING FROM BUILDING STRUCTURE

- A. The following are guidelines for supporting mechanical, electrical, plumbing, or architectural items from the building structure. Coordinate load requirements from other contractors so no combination of loads exceeds the limitations provided in the structural design.
- B. At both the floor and the roof, attachments may be at the upper or the lower truss chord (horizontal members at top and bottom of truss). Do not hang loads from web members (the diagonal and vertical members between chords), including the end diagonal member where the lower chord is discontinuous.
- C. Make the point of attachment at a panel point of the truss girders or joists. Panel points are the intersections of the horizontal chords with the diagonal or vertical web members.
- D. Attachments to metal decking require written approval from the Engineer. If requested, provide inserts where concrete fill occurs or stiffeners welded where roof insulation occurs.
- E. Do not weld on trusses. Use bolted or clamped-type connections.
- F. Comply with the Specifications for support and load requirements and restrictions. Coordinate and confirm support and load requirements with the Engineer. The Engineer will coordinate load requirements submitted by other contractors so combination loads do not exceed project structural design criteria limitations.
- G. Support loads hung from concrete structure with cast-in-place inserts unless drilled-in anchors are specifically shown or approved by the Engineer in writing.
- H. The above load restrictions and limitations shall apply to the substructure as well.

11.0 WARRANTY

- A. Warrant to the Owner that materials and equipment furnished will be of good quality and new unless otherwise required or permitted by the Contract Documents,

that the work will be free from defects not inherent in quality required or permitted, and that the work will conform with the requirements of the Contract Documents. Work not conforming to these requirements, including substitutions not properly approved and authorized, may be considered defective. The warranty excludes remedy for damage or defect caused by abuse, modifications not executed by the Contractor, improper or insufficient maintenance, improper operation, or normal wear and tear under normal usage.

- B. The warranty period shall be for a minimum of 24 months from the date of Substantial Completion of the work or designated portion thereof.
- C. Make repairs and replacements promptly upon receipt of written notification of defect from the Engineer or Owner. If, within a reasonable time, the Contractor has not made the necessary corrections, Owner may make the corrections or order the work to be done by a third party, and the cost of the corrections shall be paid by the Contractor.

- END -